No of Pages : 3 Course Code : 23Z301 / U / R / I

Roll No:

(To be filled in by the candidate)

PSG COLLEGE OF TECHNOLOGY, COIMBATORE - 641 004

SEMESTER EXAMINATIONS, DECEMBER 2024

BE / BTech – CSE / I&CE / ROBOTICS / INFORMATION TECH. Semester: 3

23Z301 / U / R / I LINEAR ALGEBRA

Time : 3 Hours **Maximum Marks :100**

INSTRUCTIONS:

1. Answer **ALL** questions. Each question carries 20 Marks.
2. Subdivisions (a)(i) and (a)(ii) carries 2 marks each, subdivision (b) carries 6 marks each and subdivision (c) carries 10 marks each.
3. Course Outcome : Table

Qn.1	CO.1	Qn.2	CO.2	Qn.3	CO.3	Qn.4	CO.4	Qn.5	CO.5

COs	At the end of learning this course, students will be able to
CO1	comprehend vector spaces and subspaces, find row space, column space and null space, identify the bases, dimension, rank and nullity of a vector space.
CO2	identify linear transformations of finite dimensional vector spaces, apply principles of matrix algebra to linear transformations and relate matrix transformations to geometric illustrations
CO3	compute inner products, create orthogonal and orthonormal bases: Gram-Schmidt process, use orthogonal and orthonormal bases to solve application problems.
CO4	compute eigenvalues and eigenvectors of a square matrix, use them to diagonalise and orthogonally diagonalise a matrix, apply properties of eigenvalues and orthogonality to analyze computational problems.
CO5	apply eigen to matrix decomposition and optimization problems.

			Question	(BTL)
1.	a)	i)	Identify the space spanned (I) by the vector (1, 0, 0) and (II) by the vectors (1, 0, 0) and (0, 1, 0).	(L2)
		ii)	Are the following statements true? Why or why not?	(L2)
			(SI) : If $\{v_1\}$ is a vector in a vector space V, then it is linearly independent.	
			(SII) : If v_2 is not a scalar multiple of v_1, then $S = \{v_1, v_2\}$ is linearly independent always.	
	b)		Given $V = \{(x, y) ; x > 0, y > 0\}$ with the operations defined as $(x_1, y_1) + (x_2, y_2) = (x_1 + x_2, y_2)$ and $k(x_1, y_1) = (x_1, y_1)$. Verify the vector space axioms. Is V a vector space? If not, list the axioms that fail to satisfy.	(L4)
	c)	i)	Find the bases and dimensions of the row space, column space and null space of the matrix A given below.	(L3)

$v_2 \neq \alpha v_1$

$\alpha v_1 - v_2 \neq 0$.

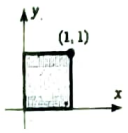
$$A = \begin{bmatrix} 2 & 4 & -3 & -6 \\ 7 & 14 & -6 & -3 \\ -2 & -4 & 1 & -2 \\ 2 & 4 & -2 & -2 \end{bmatrix}$$

(OR)

ii) Consider the bases $B = \{u_1, u_2, u_3\}$ and $B' = \{v_1, v_2, v_3\}$ for R^3, where $u_1 = (-3, 0, -3)$, $u_2 = (-3, 2, -1)$, $u_3 = (1, 6, -1)$; $v_1 = (3, 1, -5)$, $v_2 = (1, 1, -3)$, $v_3 = (-1, 0, 2)$. Find the transition matrix $B' \rightarrow B$. Also compute the coordinate vector $[w]_B$ where $w = (1, 2, 3)$.

2. a) i) Let $T: P_2 \rightarrow P_3$ be a linear transformation defined by $T(p(x)) = xp(x)$. What is the range and kernel of T? (L1)

ii) Obtain the matrix operator for the shear in the y direction given by $T: R^2 \rightarrow R^2, T(x, y) = (x, y + 2x)$, Also sketch the image of the figure given below with respect to the shear operator. (L2)



b) Consider $S = \{v_1, v_2\}$ for R^2 where $v_1 = (1,1), v_2 = (1,0)$. Let $T: R^2 \rightarrow R^2$ be the linear operator for which $T(v_1) = (1, -2)$ and $T(v_2) = (-4,1)$. Find a formula for $T(x_1, x_2)$ and use the formula to find $T(2, 2)$. (L3)

c) Let $T: R^3 \rightarrow R^3$ be the linear operator defined by $T(x_1, x_2, x_3) = (x_1 - x_2, x_2 - x_1, x_1 - x_3)$. Find the matrix for T with respect to the basis $B = \{v_1, v_2, v_3\}$ where $v_1 = (1,0,1)$, $v_2 = (0, 1, 1)$, $v_3 = (1,1,0)$. Also verify that $[T]_B[x]_B = [T(x)]_B$ holds for every vector x in R^3. (L5)

3. a) i) Find x such that the set of vectors $\{u = (1, -1), v = (1, x)\}$ is orthogonal with respect to the weighted Euclidean inner product $\langle u, v \rangle = 4u_1v_1 + u_2v_2$ on R^2. (L1)

ii) Let $x = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ be the least square solution for the system $Ax = b$ with $A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{bmatrix}, b = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$, then determine the least square error. Suppose least square solutions are infinitely many then, is least square error also infinitely many? or is it unique ? (L2)

b) Apply Gram-Schmidt process to obtain the orthonormal basis vectors for the basis vectors $(1, 1, 0), (1, 2, 0), (0, 1, 2)$. (L3)

c) Find QR decomposition of the matrix $A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$. (L3)

4. a) i) For the matrix $A = \begin{pmatrix} 1 & 0 \\ 1 & 2 \end{pmatrix}$, find the eigenvalue corresponding to the eigenvector $\begin{bmatrix} 0 \\ 20 \end{bmatrix}$. (L1)

ii) For a 2x2 symmetric matrix A, if it is observed that $A\begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $A\begin{bmatrix} 0 \\ 1 \end{bmatrix} = 2\begin{bmatrix} 0 \\ 1 \end{bmatrix}$, then determine the matrix A. (L1)

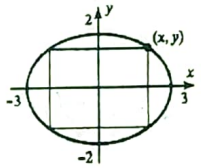
b) Find a matrix P that orthogonally diagonalizes $A = \begin{bmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{bmatrix}$. (L3)

c) Identify the conic section represented by the equation $9x^2 - 4xy + 6x^2 - 10x - 20y = 5$ by rotating axes to place the conic in standard position. Find an equation of the conic in the rotated coordinates, and also find the angle of rotation. (L4)

5. a) i) Let A be an image matrix of order 256 x 256, which obtains rank as 120 when reduced spectral decomposition is used. How much of storage space will be reduced if rank 50 approximation of A is considered? (L2)

ii) Let the eigenvalues of a 2x2 symmetric matrix A be 2 and -2 with their corresponding eigenvectors as $\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$ and $\left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$. Also let the image of $\begin{bmatrix}1\\2\end{bmatrix}$ under spectral decomposition of A be $\begin{bmatrix}3\\1\end{bmatrix}$. Apply the geometrical interpretation of spectral decomposition of A to locate the image of (1,2) graphically. (L2)

b) A rectangle is to be inscribed in the ellipse $4x^2 + 9y^2 = 36$, as shown in the figure below. Use eigenvalue method to find nonnegative values of x and y that produce the inscribed rectangle with maximum area. (L3)



c) i) Find the singular value decomposition of $A=\begin{bmatrix}1 & 0\\1 & 1\\-1 & 1\end{bmatrix}$. Also express the reduced singular value decomposition of A. (L5)

(OR)

ii) Find the spectral decomposition of $A = \begin{pmatrix}-2 & 6\\6 & -2\end{pmatrix}$. Also verify the image of the vector $x = \begin{bmatrix}0\\1\end{bmatrix}$ of Ax matches with that of spectral decomposition of A.

CO : COURSE OUTCOME; BTL : BLOOM'S TAXONOMY LEVEL

/END/

FD/JU